Sparsh Bandi | CH24B100 | Indian Institute of Technology Madras

EDUCATION AND SCHOLASTIC ACHIEVEMENTS

Program	Stream	Institution	CGPA	Year
B. Tech	Chemical Engineering	Indian Institute of Technology Madras	7.74	2028
Class XII	CBSE	Delhi Public School, Bangalore South	91.8%	2024
Class X	CBSE	Delhi Public School, Bangalore South	97.8%	2022

- Secured **AIR 3861** out of **1.8 Lakh+ qualified** candidates in **JEE Advanced '24** and **AIR 3723** (**99.78%**) out of **16 Lakh+** candidates in **JEE Mains '24**
- Scored **328/390** in **BITSAT**, placing in the top **0.05%** of candidates, achieved **95%** in PCMC subjects in Class XII and **97.8%** in Class X
- Cleared **FCC HAM Radio Technician** exam (**35 MCQs, 74%+ pass mark**), earning a **US** amateur radio operator license in **RF communication.**

COURSEWORK

Engineering Mechanics	Probability, Stochastic Processes & Statistics	Programming in Data Structures and Algorithms
Signals and Systems	Multivariate Data Analysis for Process Modelling	Fluid and Particle Mechanics
Innovation and Entrepreneurship	Fundamentals of Thermodynamics	Fundamentals of Heat and Mass Transfer

Software/Tools/Platforms	Matlab, Github, Canva, LaTeX, Apollo, AutoCAD, KiCAD, X-FOIL, Simulink, Pixhawk, ArduPilot, Arduino, Ansys(CFD)
Languages	Python, C++(Intermediate), Embedded C, LaTeX, Matlab , ROS2
Core Competencies	UAV Design, Flight Computer Development, PCB Design, YOLOv11 Object Detection, Autonomous Systems, Computational Aerodynamics, Telemetry (ExpressLRS, MAVLink), SRAD Avionics, Hybrid Propulsion Systems, ROS2

PROFESSIONAL EXPERIENCE

Valkron Dynamics
Research Intern
Jul 25' - Aug 25'

Early-stage UAV startup building real-time autonomous tracking systems using computer vision and embedded guidance.

- Optimized real-time **YOLO** object detection pipeline, reducing inference latency by **30%** for autonomous UAV tracking
- Engineered **high-performance UAV aerodynamic models**, maximizing structural integrity for reliable autonomous flight
- Implemented **PID-based guidance logic** in **ROS2**, ensuring robust performance across **50+** autonomous aerial missions

Abhyuday
Avionics and Electronics
Sep '25 - Present

Sole Avionics member in IIT Madras's 6-person IREC delegation; built full SRAD avionics for a 10,000 ft COTS rocket.

- Represented IIT Madras as sole Avionics member at **IREC USA** (**10,000 ft COTS** category) among **143 global teams**
- Engineered full **SRAD** avionics suite for a **10,000 ft COTS** rocket; cleared **IREC Flight Safety Review** for launch clearance
- Achieved a successful launch to **8,000 ft AGL** and pitched **SRAD** systems to technical leads at **Blue Origin** and **SpaceX**

RESEARCH PROJECTS

Project Icarus Prime
Aero Club
Jun '25 - Mar '26

Stealth STOL UAV for swarm-defense missions with 2 patent filings on radar interception and onboard vision systems

- Researched **5+ papers** and engineered a **STOL stealth UAV** with **NACA 9415** for autonomous swarm-defense missions
- Filed **2 patents** on object detection, guidance, and visual drone interception using radar-integrated systems
- Built autonomous **YOLOv11** vision model trained on **500+ images**, achieving **0.6 confidence** against real aerial targets
- Tuned **ArduPilot** flight controller for geofenced UAV swarm-interception, enabling fully **autonomous mission execution**

LAT Aerodynamics Problem Statement
(Inter IIT Collegiate Competition)

Inter IIT problem on USB aerodynamics; built thick jet-flap interaction models validated against high-fidelity CFD.

- Built **2D Python USB** model for **S1223** airfoil; **86 mm** jet at **70% chord** with **Coanda** effect and variable density
- Optimised **6 parameters** across **50+ AoA** configs, matching **CFD** within **0.5% error** at **~100x** lower compute time
- Ran **coflow-jet** augmentation analysis achieving **CL max of 4.5**, as sole modeller in a **30-member** Inter IIT team

Attitude Control Demonstration Matlab
Prof Aravind Kumar
(Course Project - CH2013)

9-DOF MATLAB attitude estimator fusing IMU data via complementary filter with real-time cockpit visualisation.

- Coded **9-DOF IMU** attitude estimator in MATLAB fusing **gyro, accel, and mag** via **complementary filter** at **100 Hz**
- Designed **3 one-pole LPFs** (5/5/25 Hz) for accel, mag, gyro denoising with tunable **ALPHA** fusion gains per axis
- Built **30 fps artificial horizon** and RPY visualiser with **10 s sliding window**; validated on **Arduino serial IMU**

Research Paper DPCA
Course Project – CH5440
Prof. Kannan A
(Multivariate Data Analysis)

Benchmarked fault-detection algorithms for multivariate statistical process monitoring in chemical engineering.

- Synthesized **15+** papers on **Dynamic PCA** and multivariate process monitoring into a comprehensive **5,500-word** paper
- Benchmarked **DPCA** against **7** fault-detection methods (PCA, KPCA, CVA, ICA, SFA) on the **Tennessee Eastman Process**
- Implemented **DPCA** from scratch in **Python** with control-chart visualizations across **53 variables**, 21 fault scenarios

COMPETITIONS

Shaastra Aerial Robotics Challenge

Autonomous drone challenge requiring full build and flight within a 1-week sprint at Shaastra, IIT Madras.

- Designed and piloted **2 fully autonomous drones** within a **1-week** sprint, advancing to **Round 2** of the competition
- Tuned **ArduPilot** flight controls and configured a custom **geofence** for the test-site, enabling safe autonomous flight

Boeing National Aeromodelling Competition

National aeromodelling competition, fly 30 seconds with max payload then execute a mid-flight payload drop.

- Built **5+ CNC-machined 1.5m balsa** airframes with **850 KV** brushless motor on **4S**, achieving **T/W of 1.6x** vs **1.3x** min
- Engineered **onboard electronics** for mid-flight release of **460g payload** (10 golf balls) after a **30-second** powered cruise

Signal - Quest
Finalist

Competition testing signal-processing circuit design and mathematical equation-solving under timed conditions.

- Advanced to finals from **200+** participants by clearing an **advanced-level qualifying test**, securing a **top-15** finish
- Built and tuned a **signal-processing circuit** in the finals to produce clean, **minimal-noise** output on an oscilloscope

POSITIONS OF RESPONSIBILITY

Aero Club Coordinator
Sept'25 - Mar'26

Led end-to-end aircraft manufacturing and piloting operations at IIT Madras Aero Club for national competitions.

- Designed and manufactured **7+** fixed-wing **RC aircraft** and **4+ quadcopters**, leading full build cycles for the club
- Built piloting proficiency in fixed-wing and multi-rotor UAVs via **ArduPilot** simulators and manual flight training
- Represented IIT Madras at the **Boeing National Aeromodelling Championship** as the club Coordinator for the season

Abhyuday Avionics Member
Sep'25 - Present

Built avionics and power systems for IIT Madras's IREC rocket, developing Flight computer and Telemetry from scratch.

- Designed **flight computer PCB** over **4 months** integrating **7+ sensors** and **15+ components** with embedded firmware
- Engineered **915/760 MHz** uplink-downlink telemetry pipeline streaming live **AGL altitude**, velocity, and heading
- Engineered **SRAD ground station** automating hybrid rocket fill sequences via **LoRa** with **7 km** bidirectional link
- Coded **Python GUI** to parse live **NMEA packets** from Featherweight GPS for real-time **3D rocket trajectory** display
- Executed **2** hybrid engine tests at Thaiyur achieving **3.3 kN** peak thrust and **diamond shock structure** formation

EXTRACURRICULAR ACTIVITIES

Sports

Tennis

- **AITA**-ranked athlete in top **10%** at TopSpin Tennis Academy, Bangalore; peak **UTR of 7+** with **3+** tournament wins
- Secured **4+ TNTA** state-level match wins and won the **Inter-Hostel Gold Medal** in Tennis (2025) at the institute

Basketball

- Clinched **IIT Madras Schroeters Gold Medal** in 2026 and reached the **Semifinals** in 2025 as a consistent performer
- Secured **Bronze Medal** as Vice-Captain of the **Inter-Department League**, leading the team in key competitive situations